

Melika Baghi

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Atlanta, GA

Summary

Ph.D. Candidate in Industrial & Systems Engineering at Georgia Tech with an M.S. in Statistics. Work covers experiment design, ETL and feature pipelines, predictive modeling and statistical validation, through to deployed models. Seeking data scientist and applied machine learning roles.

Technical Skills

Programming & data: Python, SQL, PySpark, R, pandas, NumPy, SciPy, Airflow, MLflow

Machine learning: scikit-learn, XGBoost, PyTorch, ensemble methods, deep learning, representation and multimodal learning

Statistics: experiment design, hypothesis testing, nonparametric methods, calibration, uncertainty quantification, causal and time-series modeling

Delivery: ETL pipelines, model tracking, dashboards (Power BI), Git, Linux, GPU/HPC

Experience

Graduate Researcher — Georgia Institute of Technology

Aug 2023 – Present

H. Milton Stewart School of Industrial & Systems Engineering · Advisors: K. Paynabar, X. Liu

- Develop machine learning and statistical methods for high-dimensional data, end to end from problem formulation through validation, reproducible code and publication.
- Built a constrained gradient-boosting estimator that keeps predictions valid during fitting rather than correcting them afterwards, cutting reconstruction error $5\times$ against interpolation baselines; released as an installable open-source package.
- Built a tensor-structured surrogate that reduced interpolation cost $35.6\times$ – $564\times$ and prediction time $4\times$ – $108\times$ while matching baseline accuracy within 0.35 percentage points.

Head Teaching Assistant — ISyE 6525: High-Dimensional Data Analytics

Jan 2024 – Present

Georgia Institute of Technology

- Lead the TA team for a graduate course of 150+ students per semester on dimensionality reduction, sparse modeling and high-dimensional inference. Named **Outstanding Online TA**, 2025.

Course Instructor — Artificial Intelligence Laboratory

Aug 2022 – Aug 2023

Royan Institute, Tehran

- Designed and taught a project-based applied AI curriculum, supervising student projects from problem framing through evaluation.

Selected Projects

- **Hospital capacity and patient flow** — hybrid agent-based and discrete-event simulation of a hospital serving two patient populations. Cut the tourist queue wait **82%** (13.7 to 2.4 days) and the admission queue **43%** (25.9 to 14.8 days, 30 replications, $p < 0.001$), with a 256-run fractional factorial design over 16 operational factors. *AnyLogic, Python*
- **Predictive maintenance at scale** — end-to-end pipeline for sensor-based fault detection. ARIMA and LSTM forecasting with ensemble anomaly classifiers over multivariate telemetry gave over **15 hours** of advance warning; deployed on Spark for low-latency inference. *Python, PySpark*
- **End-to-end ML pipelines** — CRISP-DM workflows over simulation, sensor and transactional data. ETL in PySpark and SQL, model tracking automated with Airflow and MLflow, results delivered as dashboards in Python, R and Power BI.
- **Clinical risk from EHR** — adverse drug event prediction from longitudinal records, compared across regularized regression, tree ensembles and neural models, with calibration and fairness checks across demographic groups. *Python*
- **Credit risk and A/B-style testing** — ensemble classifiers for loan default with calibration analysis; separately, a cross-national nonparametric study of internet access and female labour outcomes across 92 countries using Mann–Whitney tests with Bonferroni correction. *Python*

Education

Ph.D., Industrial & Systems Engineering (System Informatics & Control), Georgia Institute of Technology Aug 2023 – Present

M.S., Statistics, Georgia Institute of Technology

Aug 2023 – May 2026

M.S., Industrial Engineering — Healthcare Systems, Amirkabir University of Technology; GPA 4.0/4.0

2021 – 2023

B.S., Industrial Engineering (GPA 3.91/4.0) and **B.S., Aerospace Engineering** (GPA 3.79/4.0), Amirkabir University of Technology

2016 – 2021

Selected Publications

Four preprints and manuscripts under review, including [arXiv:2605.04130](#) (constrained gradient boosting), [arXiv:2608.15520](#) (adaptive modality acquisition), [arXiv:2605.19139](#) (hospital simulation) and [arXiv:2609.01761](#) (delayed bandits). Full list: [Google Scholar](#).